

GED Math: Number Sense and Problem Solving

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Lesson 1: Understanding Numbers

- Types of numbers: natural, whole, integers, rational, irrational.
- Prime and composite numbers.
- Factors and multiples.
- Greatest Common Factor (GCF) and Least Common Multiple (LCM).

Example 1: Classifying Numbers

Example: Classify the number $\sqrt{16}$.

Solution:

- $\sqrt{16} = 4$, which is a whole number.
- Therefore, it is natural, whole, integer, and rational.

Example 2: Prime Factorization

Example: Find the prime factorization of 84.

Solution:

$$84 = 2 \times 42 = 2 \times 2 \times 21 = 2^2 \times 3 \times 7$$

Lesson 2: Fractions and Decimals

- Converting between fractions, decimals, and percentages.
- Simplifying fractions.
- Adding, subtracting, multiplying, and dividing fractions.

Example 3: Simplifying a Fraction

Simplify $\frac{18}{24}$.

Solution:

$$\text{GCF}(18, 24) = 6, \quad \frac{18}{24} = \frac{3}{4}$$

Example 4: Fraction to Decimal

Convert $\frac{7}{8}$ to a decimal.

Solution:

$$\frac{7}{8} = 0.875$$

Lesson 3: Ratios, Rates, and Proportions

- Understanding ratios and rates.
- Solving proportions.
- Applications in real-world problems.

Example 5: Solving a Proportion

Solve $\frac{3}{4} = \frac{x}{12}$.

Solution:

$$3 \times 12 = 4x \Rightarrow x = 9$$

Lesson 4: Percents

- Converting between fractions, decimals, and percents.
- Finding percentage of a number.
- Percentage increase and decrease.

Example 6: Finding a Percent

Find 25% of 80.

Solution: $0.25 \times 80 = 20$

Example 7: Percentage Change

The price increases from 50 to 65. Find the percent increase.

Solution:

$$\text{Percent change} = \frac{65 - 50}{50} \times 100\% = 30\%$$

Lesson 5: Estimation and Rounding

- Rounding to the nearest whole number, tenth, hundredth.
- Estimation in problem solving.

Example 8: Rounding

Round 67.489 to the nearest tenth.

Solution: $67.489 \approx 67.5$

Lesson 6: Order of Operations

- Use PEMDAS: Parentheses, Exponents, Multiplication/Division, Addition/Subtraction.

Example 9: Evaluating an Expression

Evaluate $6 + 3 \times (2 + 4)^2$.

Solution:

$$6 + 3 \times 6^2 = 6 + 3 \times 36 = 6 + 108 = 114$$

Lesson 7: Absolute Value

- $|x|$ is the distance from zero.
- Example: $|-5| = 5$

Example 10: Comparing Numbers

Which is greater: $|-3|$ or 2?

Solution: $|-3| = 3$, so $3 > 2$.

Example 25: Multi-step Problem

A car travels 150 miles in 3 hours. How far will it travel in 5 hours at the same rate?

Solution:

$$\text{Rate} = 150/3 = 50 \text{ mph}$$

$$\text{Distance} = 50 \times 5 = 250 \text{ miles}$$